

Prior to proceed let us consider the the product $D_p x$. Since D_p is an $n \times n$ matrix with entries in $\mathbb{Q}_p$ and x is an n -tuple with entries in $\mathbb{Q}_p$, the result $D_p x$ is an n -tuple with entries in $\mathbb{Q}_p$. Formally, if $D_p = (d_{ij})$ and $x = (x_1, x_2, \dots, x_n)$, then the i -th component of $D_p x$ is given by:

$$(D_p x)_i = \sum_{j=1}^n d_{ij} x_j \in \mathbb{Q}_p$$

The p-adic norm $|\cdot|_p$ is applied component-wise to the resulting n -tuple $D_p x$. For each component $(D_p x)_i$ of $D_p x$, we have:

$$|(D_p x)_i|_p = \frac{1}{p^{\text{ord}_p((D_p x)_i)}}$$

where $\text{ord}_p((D_p x)_i)$ is the highest power of p dividing $(D_p x)_i$.

Since the p-adic norm $|\cdot|_p$ maps elements of $\mathbb{Q}_p$ to $\mathbb{Q}_p$, applying it component-wise to an n -tuple will result in another n -tuple with entries in $\mathbb{Q}_p$. Therefore, $|D_p x|_p$ is an n -tuple with entries in $\mathbb{Q}_p$.

Thus, $\delta_p(x) = |D_p x|_p \in \mathbb{Q}_p^n$, hence, the map δ_p is well-defined.

As we have seen previous we can define the p -adic Ducci sequence the sequence

Definition 0.4 (p -adic Ducci sequence).

$$x, \delta_p(x), \delta_p^2(x) = \delta_p(\delta_p(x)) = \delta_p \circ \delta_p, \dots, \delta_p^n(x), \dots$$

with initial seed $x \in \mathbb{Q}_p^n$ respect to $D_p \in \mathbb{M}_{n \times n}(\mathbb{Q}_p)$. In this p-adic context, we define a p-adic Ducci map associated with a p -adic matrix D_p as a function that maps a p-adic vector x to $|D_p x|_p$, where $|\cdot|_p$ is applied elementwise.

Without loss of generality we can define the following For convergence to zero, we analyze when $x_k \rightarrow 0$ as $k \rightarrow \infty$.

Prior to move forward we recall the following

Remark 1. In the field of p -adic numbers $\mathbb{Q}_p$, the p -adic absolute value $|x|_p$ of a nonzero element x is defined as:

$$|x|_p = p^{-v_p(x)},$$

where $v_p(x)$ is the p -adic valuation of x . The valuation $v_p(x)$ is the highest power of p that divides x in $\mathbb{Q}_p$.

If $|x|_p < 1$ and $x \neq 0$, this means that $v_p(x) > 0$, so x is divisible by p . The possible values of $|x|_p$ in this case are:

$$|x|_p = p^{-k}, \quad \text{where } k \in \mathbb{N} \text{ and } k \geq 1.$$

Thus, the possible values of $|x|_p$ when $|x|_p < 1$ are:

$$\{p^{-1}, p^{-2}, p^{-3}, \dots\}.$$